Multimodal Artificial Intelligence in Enterprise Robotic Process Automation

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***Abstract*—We build a system that extracts structured order data from images and audio recordings in Vietnamese, then matches the extracted product names to an enterprise catalog. The system uses Gemini 2.5 Flash as a native multimodal model to read invoices and listen to audio directly, outputting structured JSON. For product matching, we combine dense embeddings (Vietnamese Sentence-BERT) with sparse embeddings (learned BM25) in a hybrid search over Qdrant, guided by LLM-predicted product categories. We evaluate the pipeline on Vietnamese enterprise documents.**

***Index Terms*—Multimodal Information Extraction, Robotic Process Automation, Large Language Models, Hybrid Retrieval**

I. Introduction

Many Vietnamese businesses still use handwritten invoices and phone calls to place orders. These inputs are hard to process automatically because they mix Vietnamese with English product names, use regional abbreviations, and come in different formats (images, audio).

Rule-based systems fail on this variety. Using a large language model (LLM) for every document works but costs too much at scale. Prior work has studied OCR, ASR, and retrieval separately, but no system combines them into one pipeline for Vietnamese enterprise data.

We build a pipeline with two main parts:

1) **Extraction**: Gemini 2.5 Flash processes raw image and audio files directly (no separate OCR or ASR step) and outputs structured JSON with seller, buyer, product, quantity, and date.
2) **Product matching**: Extracted product names are normalized by an LLM, classified into a category tree (L1/L2/L3), then matched against a catalog of $\sim$50k products using hybrid dense+sparse vector search in Qdrant.

Our contributions:

1) A pipeline that uses a native multimodal LLM for direct extraction from Vietnamese invoices and audio orders.
2) A product matching module combining LLM query normalization, category prediction, and hybrid retrieval (Sentence-BERT + learned BM25 via Weighted Score Fusion).
3) Evaluation on Vietnamese enterprise documents.

II. Related Work

A. Document Understanding

LayoutLM [1] encodes text with bounding-box positions for form understanding, but needs fine-tuning per layout. Donut [2] skips OCR and decodes from images directly, but is expensive and needs training data. Our system uses Gemini 2.5 Flash to process raw image bytes natively — no OCR, no fine-tuning.

B. Vietnamese ASR

Vietnamese is tonal with six tone classes and has limited training data. PhoWhisper [3] fine-tunes Whisper [4] on 800+ hours of Vietnamese speech. VietASR [5] combines HuBERT [6] with Zipformer [7]. Both struggle with mixed Vietnamese-English product names. Instead of a dedicated ASR module, we send raw audio bytes to Gemini for joint transcription and extraction, with phonetic correction rules in the prompt.

C. LLM-Based Extraction

Traditional NER (e.g., BiLSTM-CRF [8]) needs labeled data per schema. LLMs like Gemini [9] can extract structured data zero-shot from a prompt. The downside is cost and latency per document. We use the LLM for two tasks: multimodal extraction and product query normalization, both with Pydantic schema validation and batching (up to 10 queries per call).

D. Hybrid Retrieval

BM25 [10] does exact matching but misses synonyms. Sentence-BERT [11] captures semantics but confuses similar specs (e.g., 8W vs 9W). Hybrid fusion [12] combines both. We use Weighted Score Fusion in Qdrant [13]: Vietnamese Sentence-BERT (dense) + learned BM25 via fastembed (sparse), with $\alpha = 0.7$ for dense weight. An LLM-predicted category filter narrows the search space; if it returns nothing, we fall back to unfiltered search.

III. Methodology

A. Problem

Given a multimodal input $\mathcal{M} = (V, A)$ where V is an image and A is audio (either can be absent), the system does two things:

$$f_{IE} : \mathcal{M} \to \mathcal{E}, \quad f_{ER} : \mathcal{E} \to \mathcal{D} \tag{1}$$

f_{IE} extracts entities $\mathcal{E}$ (seller, buyer, product, quantity, date) from the input. f_{ER} maps product names to canonical entries in the enterprise catalog $\mathcal{D}$.

B. Architecture

The pipeline has two stages (Fig. 1):

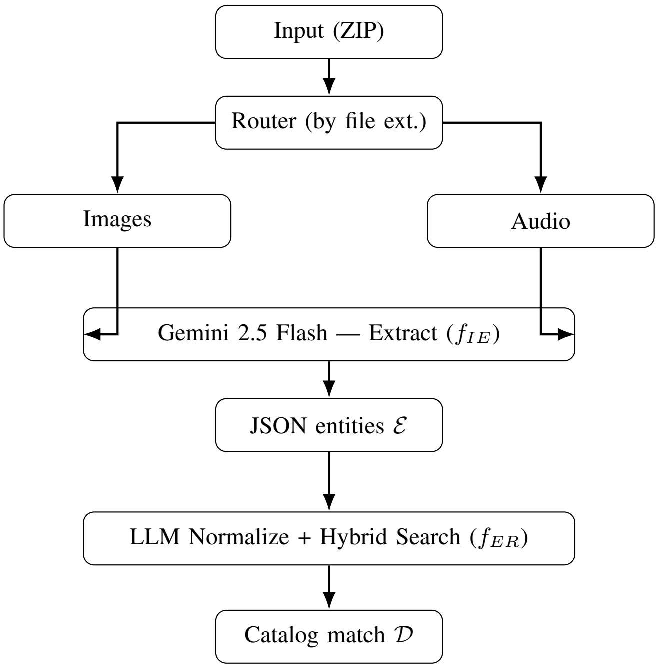

Fig. 1. Pipeline overview.

C. Stage 1: Multimodal Extraction

We send raw file bytes (image or audio) directly to Gemini 2.5 Flash with a structured prompt. The model reads the image or listens to the audio natively — no separate OCR or ASR step. The output is a Pydantic-validated JSON with fields: `seller`, `buyer`, `product_name`, `quantity`, `order_date`.

The prompt includes:

- Target field definitions.
- Phonetic correction rules for Vietnamese (e.g., "chụ" $\rightarrow$ "trụ" for LED products).
- Rules to distinguish quantity columns from unit conversion columns in handwritten invoices.
- Date normalization to `dd/mm/yyyy`.

Temperature is set to $T = 0.2$ with thinking disabled for speed.

D. Stage 2: Product Matching

Product names from Stage 1 are raw text that needs to be matched to the enterprise catalog ($\sim$50k SKUs). This is done in two steps.

1) Step 1: LLM Normalize + Classify: A second LLM call (Gemini 2.5 Flash, $T = 0.1$) processes each product name to:

1) **Normalize**: Clean the query using domain rules — remove packaging suffixes ("1T" = carton), strip price codes, fix abbreviations.
2) **Classify**: Predict the L1/L2/L3 category from the catalog taxonomy.

Rules are stored in external markdown files (4 files, $\sim$22KB total). Queries are batched up to 10 per LLM call.

2) Step 2: Hybrid Vector Search: The normalized query is searched against Qdrant using two vector types in parallel:

- **Dense**: Vietnamese Sentence-BERT (`keepitreal/vietnamese-sbert`, 768-dim, cosine).
- **Sparse**: Learned BM25 via fastembed (`Qdrant/bm25`).

Each stream returns top-100 candidates. Scores are min-max normalized per stream, then combined:

$$\text{Score}(d) = 0.7 \cdot \hat{s}_{\text{dense}}(d) + 0.3 \cdot \hat{s}_{\text{sparse}}(d) \tag{2}$$

If the LLM predicted a category, a `must` filter is applied on L1/L2/L3 metadata. If this returns no results, the system retries without the filter (automatic fallback).

IV. Experiments and Results

A. Setup

1) Data:

- $\sim$100 invoice images + $\sim$50 audio recordings with manual annotations.
- $\sim$50,000 product catalog (3-level hierarchy L1/L2/L3).

2) System: Intel i9-12900, 16 GB RAM, RTX 3090 Ti. Dense embeddings via `keepitreal/vietnamese-sbert` on GPU. Sparse embeddings via `Qdrant/bm25` (fastembed). LLM: Gemini 2.5 Flash API.

3) Baselines:

1) Legacy rule-based (regex + fixed coordinates).
2) Separate OCR + LLM (no hybrid retrieval).
3) Two-stage: PhoWhisper [3]/Whisper [4] ASR, then LLM extraction.

V. Conclusion

We built a pipeline for extracting order data from Vietnamese invoices (images) and phone recordings (audio), then matching products to an enterprise catalog.

The main ideas:

- Use Gemini 2.5 Flash as a native multimodal model — no separate OCR or ASR.
- Match products via hybrid search (Sentence-BERT + BM25) with LLM-based query normalization and category filtering.

Limitations. Small evaluation set. Depends on external LLM API (cost, latency, privacy). Only product matching is implemented — no merchant/buyer resolution yet. Rules are specific to one manufacturer.

Future work. Add merchant matching. Fine-tune a smaller on-premise model. Add reranking. Test on more datasets.

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